Jagraj Gill

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EDUCATION

Simon Fraser University, **December 2027**

Bachelor of Science in Computing Science

Relevant Coursework: Data Structures & Algorithms I, Data Structures & Algorithms II, Software Engineering, Databases

EXPERIENCE

Software Engineer Intern, Savi Finance – Toronto, ON **May 2025 – December 2025**

- Developed an AI-powered task scheduling system that ranked tasks by priority, **reducing planning time by 10%**
- Engineered shared modules across React and React Native backed by MongoDB services for a fintech SaaS platform serving **1000+ users, shortening release cycles by 15%**
- Built and integrated image upload workflow using AWS S3 and GraphQL, supporting custom image previews and improving upload reliability while **lowering image load failures by 10%**
- Refactored login UI and validation using React and custom hooks, **cutting user login errors by 45%**

Hackathon, Storm Hacks 2024 – Burnaby, BC **May 2024**

- Designed and deployed REST APIs with Node.js and Express.js to enable real-time data updates, **improving system response time by 30%**
- Directed a team during a 24-hour hackathon, coordinating tasks and project delivery under strict deadlines

PROJECTS

IoT Climate Monitoring Dashboard | C, ESP-IDF, FastAPI, SQLite, JavaScript, HTML, CSS **May 2026**

- Built an end-to-end IoT climate monitoring system integrating microcontroller firmware, I2C sensor communication, OLED display output, REST APIs, SQLite storage, and a live web dashboard
- Developed ESP-IDF firmware to collect temperature and humidity readings, update an OLED interface, and transmit structured JSON data to a FastAPI backend over Wi-Fi
- Engineered 24-hour climate tracking with SQLite and REST endpoints, enabling a JavaScript dashboard to display live readings, device status, and trend visualizations

AI Study Pal | Python, FastAPI, Gemini API, HTML, CSS, JavaScript **April 2026**

- Built an end-to-end pipeline that converts PDF documents into structured flashcard datasets by extracting, cleaning, and processing unstructured text with LLM APIs
- Designed a strict JSON schema for AI-generated flashcards, improving output consistency and enabling reliable parsing
- Implemented validation logic in Python to verify data structure, field integrity, and content quality, preventing malformed or incomplete AI responses from entering the system
- Developed an interactive web interface using HTML, CSS, and JavaScript to display flashcards with flip and navigation features, enabling real-time study from generated content

Wireless 4-DOF Robotic Arm | ESP32, C, Embedded Systems **February 2026**

- Designed and fabricated a 4-DOF robotic arm using ESP32 microcontrollers, servos, and custom 3D-printed components
- Developed embedded C software to process analog inputs and generate PWM signals for real-time actuator control
- Implemented wireless control between dual ESP32 devices, enabling synchronized motion through real-time transmission
- Engineered modular control architecture supporting future inverse kinematics and sensor integration

Investment Tracker | Java, Swing, JFreeChart **May 2025**

- Built a Java desktop app with a Swing-based GUI to visualize investment growth using side-by-side JFreeChart graphs
- Engineered portfolio logic to calculate asset performance using spot prices, quantities, and historical purchase data
- Improved reliability through exception handling and input validation, ensuring consistent behavior for dynamic entries

SKILLS

Languages: C, C++, Embedded C, Python, Java, JavaScript, x86 Assembly
Web & Backend: React, HTML5, CSS, Node.js, Express.js, GraphQL, FastAPI
Databases & Tools: MongoDB, MySQL, SQLite, AWS S3, Git, Linux
Embedded Systems: Microcontrollers, RTOS, Hardware Interfacing, I2C, UART, TCP, SSH